SPORTS & ARTHROSCOPY · NEET-SS / FET / INI-SS

Elbow in throwers

1. Throwing biomechanics

- **Phases of throwing: wind-up**, **early cocking**, **late cocking** (maximum shoulder abduction and **external rotation**), **acceleration** (**most rapid motion in sport: shoulder internal rotation up to about 7,000 degrees/second**), **deceleration** (eccentric posterior cuff load), **follow-through**.
- **Elbow valgus load peaks in late cocking and early acceleration**: about **64 Nm of valgus torque** at the elbow, which **exceeds the ultimate tensile strength of the UCL (about 32-34 Nm)**: the remainder is resisted by the **flexor-pronator mass, the radiocapitellar joint and the posteromedial olecranon**.
- **Valgus extension overload (VEO):** valgus + rapid extension creates:
  - **Medial tension**: **UCL injury**, **medial epicondyle apophysitis/avulsion**, **flexor-pronator strain**, **ulnar neuritis**.
  - **Lateral compression**: **radiocapitellar overload: OCD of the capitellum**, **Panner disease**, radial head changes, loose bodies.
  - **Posterior shear (posteromedial impingement)**: **olecranon osteophytes (posteromedial)**, **olecranon stress fractures**, loose bodies.
- **Kinetic chain: legs and trunk generate about 50% of velocity/force**; **scapular dyskinesis**, **GIRD** (glenohumeral internal rotation deficit over about 20 degrees or total arc loss), and poor mechanics increase elbow and shoulder loads.

2. Ulnar collateral ligament (UCL / medial collateral ligament)

- **Three bundles:**

  - **Anterior bundle (the most important: primary restraint to valgus from about 20-120 degrees)**: from the **anteroinferior surface of the medial epicondyle** to the **sublime tubercle** of the coronoid; **anterior band tight in extension, posterior band tight in flexion**.
  - **Posterior bundle** (floor of the cubital tunnel; tight in flexion: **contracture causes loss of flexion**).
  - **Transverse bundle (Cooper ligament)**: no stabilising role.
- **Valgus stability: in full extension: the anterior capsule, UCL and bony articulation each contribute about one-third**; **at 90 degrees: the UCL (anterior bundle) provides about 55-70%**; **radial head = secondary valgus stabiliser** (important if UCL deficient).

Injury

- **Chronic attenuation** (repetitive microtrauma) or **acute rupture** ("pop" during throwing); **medial elbow pain in late cocking/early acceleration**, **loss of velocity and control**, **ulnar nerve paraesthesia** (about 40%).
- **Examination tests:**
  - **Moving valgus stress test (O'Driscoll)**: **most sensitive (about 100%)**: valgus stress while rapidly extending from full flexion: **pain between about 70 and 120 degrees ("shear angle")**,
  - **Milking manoeuvre** (pull on the thumb with shoulder abducted, elbow flexed at 90 degrees),
  - **Valgus stress test at about 30 degrees of flexion** (unlocks the olecranon).
- **Imaging: stress radiographs** (medial joint opening over about 3 mm side-to-side difference), **MR arthrography** (**"T sign"**: **undersurface (partial) tear at the sublime tubercle** with dye tracking distally along the ulna), **dynamic ultrasound**.
- **Treatment:**

- **Non-operative:** rest from throwing (about 6-12 weeks), **rehabilitation (flexor-pronator strengthening, kinetic chain)**, **PRP (partial tears)**; good for **partial tears and non-throwers**.
- **Reconstruction ("Tommy John" surgery; Jobe 1974):**
- **Graft: palmaris longus (ipsilateral; absent in about 15%)**, **gracilis**, plantaris.
- **Original Jobe technique: flexor-pronator detachment + submuscular ulnar nerve transposition (ulnar neuropathy in about 20%+)**.
- **Modified Jobe** (**muscle-splitting approach**, no routine transposition).
- **Docking technique (Altchek)**: **single humeral tunnel with the graft ends docked**; **higher return-to-play rates**.
- **DANE TJ** (hybrid).
- **Return to play about 80-90%**, after about **12-18 months** (pitchers).
- **UCL repair with internal brace (suture tape augmentation)**: **proximal or distal avulsions with good tissue quality in young athletes**; **faster return (about 6-7 months)**.

- **Complications: ulnar nerve neuritis (commonest)**, medial antebrachial cutaneous nerve injury, **medial epicondyle fracture (tunnel)**, graft failure, stiffness.

3. Valgus extension overload and posteromedial impingement

- **Posteromedial olecranon osteophytes** impinge on the **olecranon fossa** in extension: **posterior elbow pain at ball release**, **loss of terminal extension**, **loose bodies**, **chondromalacia of the posteromedial trochlea ("kissing lesion")**.
- **Test: valgus extension overload test** (repeated forced extension with valgus: posteromedial pain).

- **Treatment:** rehab, **arthroscopic osteophyte excision** (**excise only the osteophyte, avoid over-resection of the normal olecranon: increases UCL strain**); **always assess the UCL** (VEO and UCL insufficiency coexist).
- **Olecranon stress fractures** (**transverse or oblique**; **triceps traction** or impingement); **ORIF with compression screw** if not healing.

4. Lateral compression: capitellum lesions

	Panner disease	Osteochondritis dissecans (OCD) of the capitellum
Age	**Under 10 years (about 5-10)**	**Adolescents (about 12-16)**
Pathology	**Osteochondrosis**: **fragmentation of the whole ossific nucleus of the capitellum**	**Focal subchondral lesion**, anterolateral capitellum; **articular cartilage involved**
Prognosis	**Self-limiting**: reossifies	**Variable**: loose bodies, arthritis
Treatment	**Rest, avoid throwing, symptomatic**; no surgery	**Stable** (open capitellar physis, good ROM, intact cartilage): **rest 3-6 months**; **unstable/failed**: **arthroscopic debridement + microfracture** (small lesions), **fixation**, **osteochondral autograft (OATS from the knee)** for large lesions (**over about 50% of capitellar width or involving the lateral margin**)

- **Typical patients: baseball pitchers, gymnasts** (upper limb weight bearing).
- **Pseudodefect of the capitellum** (normal posterior non-articular groove: do not mistake for OCD).

5. Medial epicondyle problems in skeletally immature throwers

- **"Little League elbow"** (**medial epicondyle apophysitis**): **medial elbow pain**, **widening/fragmentation of the medial epicondyle apophysis**; **rest from throwing (about 4-6 weeks)** and gradual return with **pitch count limits** (**pitch-count guidelines by age**, avoid pitching year-round, avoid curveballs in young pitchers (controversial), no pitching with fatigue).
- **Medial epicondyle avulsion fracture: displacement over 5 mm (or 2 mm in throwers/dominant arm)**, **incarceration in the joint (absolute indication)**, **ulnar nerve dysfunction**, **valgus instability**: **ORIF (screw)**.

6. Other elbow conditions in athletes

- **Lateral epicondylitis (tennis elbow): ECRB origin (angiofibroblastic tendinosis: Nirschl)**; **ages 35-55**; **Cozen test (resisted wrist extension)**, **Mill test**, **Maudsley test (resisted middle finger extension)**. **Self-limiting (about 80-90% resolve within 1 year)**; **physiotherapy (eccentric loading)**, **counterforce brace**; **steroid injection: short-term benefit but worse outcomes at 1 year than wait-and-see (Bisset, Coombes)**; **PRP** (mixed evidence); **surgery (open, percutaneous or arthroscopic ECRB release/debridement)** after **6-12 months** of failed conservative care. **Rule out radial tunnel syndrome and posterolateral rotatory instability (PLRI)** (**iatrogenic LUCL injury during lateral release/debridement**: stay anterior to the radial head equator).
- **Medial epicondylitis (golfer's elbow): pronator teres and FCR origin**; **ulnar neuritis coexists in about 20-60%**; similar treatment.
- **Distal biceps tendon rupture: eccentric load on a flexed elbow**, **dominant arm of men aged 40-60**, **smokers, anabolic steroids**; **hook test (O'Driscoll: most sensitive and specific)**, **reverse Popeye**, **loss of supination strength (about 40-50%) and flexion**

strength (about 30%). **Early repair (within 2-3 weeks)** via **single-incision (anchors, cortical button: higher LABCN (lateral antebrachial cutaneous nerve) injury: commonest complication)** or **two-incision (Boyd-Anderson: heterotopic ossification/radioulnar synostosis)**. **Chronic (over about 6 weeks): reconstruction with graft** (Achilles allograft). **Lacertus fibrosus intact** may limit retraction.

- **Distal triceps rupture:** rare; **anabolic steroids, olecranon bursitis, renal failure, local steroid injection**; **"fleck sign" (olecranon avulsion)**; repair.
- **PLRI (posterolateral rotatory instability): LUCL (lateral ulnar collateral ligament) insufficiency**: **after elbow dislocation**, iatrogenic, cubitus varus; **lateral pivot-shift test (O'Driscoll: supination, valgus, axial load in extension: subluxation, reduces with flexion at about 40 degrees)** (best under anaesthesia), **chair push-up test**, **table-top relocation test**; **LUCL reconstruction (docking, palmaris longus)**.
- **Elbow arthroscopy portals:**
  - **Proximal anteromedial portal (about 2 cm proximal to the medial epicondyle, anterior to the intermuscular septum)**: **medial antebrachial cutaneous nerve**; **ulnar nerve** (posterior to the septum: **check for ulnar nerve subluxation/previous transposition before medial portals**),
  - **Anterolateral portal**: **radial nerve (PIN)** (the **closest major nerve: about 4-7 mm; distended joint and flexion move it away**),
  - **Proximal anterolateral portal** (safer from the radial nerve than distal anterolateral),
  - **Direct lateral ("soft spot") portal**: posterior antebrachial cutaneous nerve,
  - **Posterior (posterocentral, posterolateral)**.

    - **Most common complication of elbow arthroscopy: transient nerve palsy** (radial/ulnar/median).
- **Elbow stiffness (post-traumatic):** functional arc **30-130 degrees (Morrey)** (**modern tasks need about 27-149 degrees**); **arthroscopic or open (column procedure) release**; **ulnar nerve decompression if flexion under about 90-100 degrees pre-op** (risk of post-op ulnar neuritis when flexion improves).

> Exam pearl: **Valgus torque in late cocking/early acceleration exceeds the UCL: flexor-pronator, radiocapitellar joint and olecranon share load. VEO = medial tension (UCL), lateral compression (capitellar OCD/Panner), posterior shear (posteromedial osteophytes).**

> Exam pearl: **UCL anterior bundle = primary valgus restraint 20-120 degrees; moving valgus stress test most sensitive (pain 70-120 degrees); MR arthrography T sign. Reconstruction (palmaris longus, docking), ulnar neuritis = commonest complication. Internal-brace repair for avulsions in young throwers.**

> Exam pearl: **Panner (under 10, whole capitellum, self-limiting) vs capitellar OCD (adolescents, focal, may need surgery: OATS for large/lateral-margin lesions). Medial epicondyle fracture: incarcerated fragment = absolute surgery; throwers fix over 2 mm. Distal biceps: hook test; single-incision = LABCN injury; two-incision = synostosis.**

Written by Claude for Ortho preSSaleel · updated 2026-11-14. Check doubtful points in a standard textbook.